

Regression_Assignment

1. Problem Statement:

The given is a dataset of a number of people claimed for the insurance. We need to predict the insurance charges based on the different parameters provided.

2. Info about dataset:

The dataset contains the age, bmi, gender, smoker/not columns. The total number of rows and columns are

*) 1338 rows

*) 6 columns

3. Data Preprocessing:

Yes the dataset contains the nominal columns and it has to be converted into numerical data.

4. Developing a good model:

I have tried all the machine learning algorithm starting from Multiple linear regression, Support vector machine, Decision tree, Random forest. The best model for this dataset is Random forest.

5. R2_Scores of all models:

S.no	Algorithm	R2_Score
1	Multiple linear Regression	0.78
2	SVM	-0.08
3	Decision Tree	0.67
4	Random forest	0.85

6. Best Model:

From the r^2 _score value of the above algorithms, Random forest is the best model for this dataset with r _score value of 0.85